Cloud Infrastructure and Distributed Systems

Modern applications rely heavily on cloud infrastructure to provide scalability, reliability, and fault tolerance. Cloud platforms such as Amazon Web Services allow developers to deploy applications globally without managing physical hardware.

Virtual Machines and Containers

Virtual machines emulate entire operating systems and allow multiple isolated systems to run on the same physical hardware. Containers are more lightweight because they share the host operating system kernel while isolating applications and dependencies.

Docker is a popular containerization platform used to package applications consistently across development and production environments. Kubernetes is an orchestration platform that automates deployment, scaling, and management of containers.

Load Balancing

Load balancers distribute incoming traffic across multiple servers. This prevents overload on individual machines and improves application reliability. Elastic Load Balancers are commonly used in AWS environments.

Microservices

Microservices architecture divides applications into smaller independent services. Each service can scale independently and communicate through APIs or message queues.

Event-Driven Systems

Event-driven architectures use asynchronous communication between services. Services publish events while other services consume them. Message queues such as AWS SQS help decouple services and improve resilience.

Serverless Computing

Serverless computing allows developers to run code without provisioning servers. AWS Lambda automatically scales functions based on incoming events. This model is cost-efficient because users only pay for execution time.

Caching

Caching improves application performance by storing frequently accessed data in memory. Redis is commonly used to reduce database queries and improve response times.

Database Scaling

Applications often use read replicas and partitioning strategies to scale databases. PostgreSQL supports replication for high availability and improved performance.

Monitoring and Logging

Monitoring systems collect metrics and logs from applications and infrastructure. CloudWatch and Grafana are commonly used to visualize performance and detect failures in production systems.

CI/CD Pipelines

Continuous integration and continuous deployment automate software delivery. GitHub Actions can automatically test and deploy applications whenever new code is pushed to a repository.

Security

Modern applications require authentication, authorization, encryption, and rate limiting. JWT authentication is commonly used in APIs to securely identify users.

Conclusion

Cloud-native applications combine distributed systems principles, scalable infrastructure, automation, and observability to support reliable production systems serving large numbers of users.